

Database Management Systems

CPS 3740

Lecture 02

Computer Science, Kean University

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Outline

- Continue last lecture
- Code Samurai and workshop
- Chapter 6
 - Exercises on Database
- Do a lot of exercises on command line
- Homework & lab exercises
 - Start working on them ASAP
- Test your accounts (Application and database).
- Evaluation test

Code Samurai and workshop

- Workshop schedule
 - Join clubs: ACM, ACM-W, Cybersecurity
 - <https://kean.campuslabs.com/engage/organizations>
 - <https://kean.campuslabs.com/engage/organization/acm>
 - Many activities
 - <http://yoda.kean.edu/~samurai/workshop.php>
- Code Samurai
 - A peer-mentoring program for CS/IT students
 - Bring your questions
 - They won't give you answers directly on assignments
 - Check schedule online at <http://yoda.kean.edu/~samurai/>

Structure Query Language

- **Data definition language (DDL)**
 - Define tables, data types, structures and data constraints.
 - All specifications are stored in the database.
 - Affect at table and structure level
 - Commands: CREATE, ALTER, DROP
- **Data manipulation language (DML)**
 - General enquiry facility (query language) of the data.
 - Affect at data level
 - Commands: SELECT, UPDATE, INSERT, DELETE
- **Data Control language (DCL)**
 - Controlling access to database for security and integrity
 - Commands: GRANT, REVOKE

Database Languages

- Data Definition Language (DDL)
 - Allows the DBA or user to describe and name entities, attributes, and relationships required for the application
 - plus any associated integrity and security constraints.
 - Structure/Table level
 - CREATE, ALTER, DROP
- Create your account on database server
 - Class website
 - Understand the create statement
 - You only need one account to access different database instances.

Database Languages

- Data Control Language (DCL)
 - Allows database administrators to configure security access to relational databases.
 - Give privilege, permissions to users on data, table, procedures, database
 - GRANT, REVOKE
- Get privilege by yourself on class website
 - Database: dreamhome, TECH3740, TECH3740_2020F
- Type the following statement under MySql prompt
mysql> use dreamhome;
- What happened?
- Database name is case sensitive.

Database Languages

- Data Manipulation Language (DML)
 - Provides basic **data** manipulation operations on data held in the database.
 - **SELECT, INSERT, DELETE, UPDATE**
- Procedural DML (will learn later)
 - allows user to tell system exactly how to manipulate data.
- Non-Procedural DML (command line)
 - allows user to state what data is needed rather than how it is to be retrieved.
- Type the following statement under MySql prompt
mysql>select * from Branch;
- **Table name is case sensitive.**

Chapter 6

**SQL – Structure Query Language
Data Manipulation**

Alternative Terminology for Relational Model

Formal terms	Alternative 1	Alternative 2
Relation	Table	File
Tuple	Row	Record
Attribute	Column	Field

- A Relation is a Table
- A Table is not necessary a Relation

Basic SQL

How does a user (human & software) communicate with database?

- **SQL (Structured Query Language)**
 - Statements for data definitions, queries, and updates (both DDL and DML)
 - Considered one of the major reasons for the commercial success of relational databases
- Each statement in SQL ends with a **semicolon**
- **Different DBMS might have different syntax.**
 - Some textbook SQL statements are NOT in MySQL.

Chapter 6 - Objectives

- Purpose and importance of SQL.
- How to retrieve data from database using **SELECT** and:
 - Use compound **WHERE** conditions.
 - Sort query results using **ORDER BY**.
 - Use aggregate functions.
 - Group data using **GROUP BY** and **HAVING**.
 - Use **subqueries**.
- **Join** tables together.
- Perform **set** operations (**UNION**, **INTERSECT**, **EXCEPT**).
- How to **update** database using **INSERT**, **UPDATE**, and **DELETE**.

SELECT Statement

```
SELECT [DISTINCT | ALL]  
    { * | [columnExpression [AS newName]] [,...]  
    }  
  
FROM          TableName [alias] [, ...]  
[WHERE         condition]  
[GROUP BY     columnList]  
[HAVING       condition]  
[ORDER BY     columnList]
```

The order of the reserved words are very important!

Writing SQL Commands

- SQL statement consists of **reserved words** and *user-defined words*.
 - Reserved words are a fixed part of SQL and must be spelt exactly as required and cannot be split across lines.
 - **User-defined** words are made up by user and represent names of various database objects such as **relations, columns, views**.

Writing SQL Commands

- Use extended form of BNF notation:
 - **Upper-case** letters represent **reserved** words.
 - **Lower-case** letters represent **user-defined** words.
 - **|** indicates a *choice* among alternatives.
 - **Curly braces {}** indicate a *required element*.
 - **Square brackets []** indicate an *optional element*.
 - **...** indicates *optional repetition* (0 or more).

Literals

- Literals are **constants** used in SQL statements.
- All non-numeric literals must be enclosed in **single quotes** (e.g. 'London').
- All **numeric** literals must **not be enclosed in quotes** (e.g. 650.00).

SELECT Statement

SELECT Specifies which columns are to appear in output.

FROM Specifies table(s) to be used.

WHERE **Filters** rows.

GROUP BY Forms groups of rows with same column value.

HAVING Filters groups subject to some condition.

ORDER BY Specifies the order of the output.

- **Order of the clauses cannot be changed.**
- **Only SELECT and FROM are mandatory.**

Examples of SQL Statements

- Consists of standard English words:

1) **CREATE TABLE Staff_xxxx (staffNo VARCHAR(5),
 IName VARCHAR(15),
 salary DECIMAL(7,2));**

2) **INSERT INTO Staff_xxxx VALUES ('SG16', 'Brown', 8300);**

3) **SELECT staffNo, IName, salary
 FROM Staff_xxxx
 WHERE salary > 10000;**

(note: xxxx is your email ID without @kean.edu)

Class exercises

- Textbook examples are created in **dreamhome** database instance
- **mysql> show tables;**
- For now, students has only **SELECT, CREATE** privileges.
- It is very important to do all exercises.
- You can create your own questions and verify the results by checking the tables.
- **Hands-on exercises are very important.**

Connect to the Database - 1

- Create DB account first at class website
 - Database server name: **imc.kean.edu**
 - Database name: **dreamhome**
- ```
$ mysql -u xxxx -p -h imc.kean.edu dreamhome
```
- Password is your Kean ID in digits
  - **xxxx** : Your kean email ID
  - Or, you can select the database name after login
- ```
$mysql -u xxxxx -p -h imc.kean.edu  
mysql> use dreamhome;  
mysql> show tables;
```

Connect to the database - 2

- Use UP  down  key
- List all databases you can access on the server
`mysql> show databases;`
- List all tables you can access on the particular database
`mysql> show tables;`

Connect to the database - 3

```
mysql> show columns from Products;
```

```
mysql> desc Products;
```

- Table name is case sensitive.

Field	Type	Null	Key	Default	Extra
P_Id	int(11)	NO	PRI	NULL	
Name	varchar(200)	YES		NULL	
Price	float	YES		NULL	
Quantity	int(11)	YES		NULL	
V_Id	int(11)	NO	MUL	NULL	

Connect to the database - 4

```
mysql> SELECT * from Vendors;
```

```
mysql> quit;
```

- Table Vendors has 4 columns and 6 rows.
- We will learn more SQL statements later.

V_Id	Name	Zipcode	State
1001	XXX	07101	NJ
1002	ABC	07811	CA
1003	CCC	07083	NJ
1004	YYY	07073	NJ
1005	QQQ	07073	NJ
1006	MMM	07811	NY

Wait...

- DO NOT create any tables in the class database
 - You are not able to drop any tables.
 - You need to follow requirements and naming convention to create tables.
 - You need to go to class site to get proper privileges
- ChatGPT will be used for teaching and practicing SQL problem-solving exercises.
 - Students may have different questions and answers.
 - Understand the explanations provided.
- Students should complete all the textbook examples.

Example 6.1 All Columns, All Rows

- List full details of all staff.

SELECT staffNo, fName, lName, position, sex, DOB, salary,
branchNo **FROM** dreamhome.Staff;

- Can use ***** as an abbreviation for **'all columns'**:

SELECT * FROM dreamhome.Staff;

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000.00	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000.00	B003
SG14	David	Ford	Supervisor	M	24-Mar-58	18000.00	B003
SA9	Mary	Howe	Assistant	F	19-Feb-70	9000.00	B007
SG5	Susan	Brand	Manager	F	3-Jun-40	24000.00	B003
SL41	Julie	Lee	Assistant	F	13-Jun-65	9000.00	B005

- Output column names will be the same in SELECT statement.**

Example 6.2 Specific Columns, All Rows

- Produce a list of salaries for all staff, showing only staff number, first and last names, and salary.

SELECT staffNo, fName, lName, salary **FROM** Staff;

staffNo	fName	lName	salary
SL21	John	White	30000.00
SG37	Ann	Beech	12000.00
SG14	David	Ford	18000.00
SA9	Mary	Howe	9000.00
SG5	Susan	Brand	24000.00
SL41	Julie	Lee	9000.00

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003
SL21	John	White	Manager	M	1945-10-01	30000.00	B005
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005

Class exercises – SQL

- Connect to the database on Linux terminal
- Show **Staff** table structure (desc command)
 - Know the column names, data types
- Show the first name, salary, gender for all **Staff**.

Example 6.3 Use of DISTINCT

- List the property numbers of all properties that have been viewed.

SELECT propertyNo **FROM** dreamhome.Viewing;

clientNo	propertyNo	viewDate	comment
CR56	PA14	2013-05-24	too small
CR56	PG36	2013-04-28	
CR56	PG4	2013-05-26	NULL
CR62	PA14	2013-04-14	no dining room
CR76	PG4	2013-04-20	too remote

propertyNo
PA14
PG4
PG4
PA14
PG36

duplicates

Example 6.3 Use of DISTINCT

- Use **DISTINCT** to **eliminate duplicates**:

SELECT DISTINCT propertyNo **FROM** dreamhome.Viewing;

clientNo	propertyNo	viewDate	comment
CR56	PA14	2013-05-24	too small
CR56	PG36	2013-04-28	
CR56	PG4	2013-05-26	NULL
CR62	PA14	2013-04-14	no dining room
CR76	PG4	2013-04-20	too remote

propertyNo
PA14
PG4
PG36

Class exercises - distinct

Step1: Write your queries for the following questions.

- Given the Staff table (staffNo, fName, lName, position, sex, DOB, salary, branchNo) , write and explain an SQL query to find
 - Show all the different positions in the Staff table
 - Show all the different branchNo, position in the Staff table

ChatGPT SQL exercises

Step2: Ask ChatGpt: Copy/paste the above statements into ChatGPT.

Step3: Discussion:

- Who has the correct answers?
- Execute the queries in the database, and verify the results.
- Can you explain your own queries?

Note: ChatGPT is for assistance only. It is not allowed for use during exams. Therefore, you should learn how to solve the problems on your own.

Example 6.4 Calculated Fields

- Produce list of monthly salaries for all staff, showing staff number, first/last name, and salary.

SELECT staffNo, fName, lName, salary/12

FROM dreamhome.Staff;

staffNo	fName	lName	salary/12
SL21	John	White	2500.00
SG37	Ann	Beech	1000.00
SG14	David	Ford	1500.00
SA9	Mary	Howe	750.00
SG5	Susan	Brand	2000.00
SL41	Julie	Lee	750.00

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003
SL21	John	White	Manager	M	1945-10-01	30000.00	B005
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005

Example 6.4 Calculated Fields

- To name column, use **AS** clause:

SELECT staffNo, fName, lName, salary/12 **AS** monthlySalary **FROM** dreamhome.Staff;

staffNo	fName	lName	monthlySalary
SL21	John	White	2500.00
SG37	Ann	Beech	1000.00
SG14	David	Ford	1500.00
SA9	Mary	Howe	750.00
SG5	Susan	Brand	2000.00
SL41	Julie	Lee	750.00

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003
SL21	John	White	Manager	M	1945-10-01	30000.00	B005
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005

Example 6.5 Comparison Search Condition

- List all staff with a salary greater than 10,000.

SELECT staffNo, fName, lName, position, salary
FROM dreamhome.Staff **WHERE** salary > 10000;

staffNo	fName	lName	position	salary
SL21	John	White	Manager	30000.00
SG37	Ann	Beech	Assistant	12000.00
SG14	David	Ford	Supervisor	18000.00
SG5	Susan	Brand	Manager	24000.00

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003
SL21	John	White	Manager	M	1945-10-01	30000.00	B005
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005

Example 6.6 Compound Comparison Search Condition

- List addresses of all branch offices in London or Glasgow.

SELECT *

FROM dreamhome.Branch

WHERE city = 'London' **OR** city = 'Glasgow';

branchNo	street	city	postcode
B005	22 Deer Rd	London	SW1 4EH
B003	163 Main St	Glasgow	G11 9QX
B002	56 Clover Dr	London	NW10 6EU

branchNo	street	city	postcode
B002	56 Cover Drive	London	NW10 6EU
B003	163 Main Street	Glasgow	G11 9QX
B004	32 Manse Road	Bristol	BS99 1NZ
B005	22 Deer Road	London	SW1 4EH
B007	16 Argyll Street	Aberdeen	AB2 3SU

Class exercises – where conditions

- Show all female's first names and their salary.
- Show the branchno that has a supervisor whose salary is greater than 15000.
- Show the first name and birthday for all female staff whose birthday is before 1950-01-01.
- List of monthly salaries for all male staff, showing the staff number, the first and last names, and the salary details.

Example 6.7 Range Search Condition

- List all staff with a salary between 20,000 and 30,000.
SELECT staffNo, fName, lName, position, salary
FROM dreamhome.Staff
WHERE salary **BETWEEN** 20000 **AND** 30000;
- BETWEEN test **includes the endpoints** of range.

staffNo	fName	lName	position	salary
SL21	John	White	Manager	30000.00
SG5	Susan	Brand	Manager	24000.00

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003
SL21	John	White	Manager	M	1945-10-01	30000.00	B005
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005

Example 6.7 Range Search Condition

- Also a negated version **NOT BETWEEN**.
- BETWEEN does not add much to SQL's expressive power. Could also write:

```
SELECT staffNo, fName, lName, position, salary  
FROM dreamhome.Staff  
WHERE salary >= 20000 AND salary <= 30000;
```

- Useful, though, for a range of values.

Example 6.8 Set Membership

- List all managers and supervisors.

```
SELECT staffNo, fName, lName, position  
FROM dreamhome.Staff  
WHERE position IN ('Manager', 'Supervisor');
```

staffNo	fName	lName	position
SL21	John	White	Manager
SG14	David	Ford	Supervisor
SG5	Susan	Brand	Manager

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003
SL21	John	White	Manager	M	1945-10-01	30000.00	B005
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005

Example 6.8 Set Membership

- There is a negated version (**NOT IN**).
- IN does not add much to SQL's expressive power. Could have expressed this as:

```
SELECT staffNo, fName, lName, position  
FROM dreamhome.Staff  
WHERE position='Manager' OR  
       position='Supervisor';
```

- **IN is more efficient when set contains many values**, especially results from subqueries.

Example 6.9 Pattern Matching

- Find all owners with the string 'Glasgow' in their address.

```
SELECT ownerNo, fName, lName, address, telNo
FROM dreamhome.PrivateOwner
WHERE address LIKE '%Glasgow%';
```

ownerNo	fName	lName	address	telNo
CO87	Carol	Farrel	6 Achray St, Glasgow G32 9DX	0141-357-7419
CO40	Tina	Murphy	63 Well St, Glasgow G42	0141-943-1728
CO93	Tony	Shaw	12 Park Pl, Glasgow G4 0QR	0141-225-7025

ownerNo	fName	lName	address	telNo	eMail	password
CO40	Tina	Murphy	63 Well St, Glasgow G42	0141-943-1728	tinam@hotmail.com	123
CO46	Joe	Keogh	2 Fergus Dr, Aberdeen AB2...	01224-861212	jkeogh@lhh.com	123
CO87	Carol	Farrel	6 Achray St, Glasgow G32 ...	0141-357-7419	cfarrel@gmail.com	123
CO93	Tony	Shaw	12 Park Pl, Glasgow G4 0QR	0141-225-7025	tony.shaw@ark.com	123

Example 6.9 Pattern Matching

- SQL has two special **pattern matching** symbols:
 - **%**: sequence of zero or more characters;
 - **_** (underscore): any **single** character.
- **LIKE** **'%Glasgow%'** means a sequence of characters of **any length** containing '*Glasgow*'.

Class exercises – Pattern Matching

- List the staff names whose last name ends with a “d”.
- List the staff names whose last name **does NOT** begins with a “B”.
- List the staff names whose last name **does NOT** have a “e” in the 2nd letter.
- List the staff names whose last name begins with “B” and the 3rd letter is a “e”.

Example 6.10 NULL Search Condition

- Use **dreamhome.Viewing** table
 - List details of all viewings on property PG4 where a comment has not been supplied.
 - There are 2 viewings for property PG4, one with and one without a comment.
- Have to test for **null explicitly** using special keyword **"IS NULL"**.
- **NULL is unknown, not 0, and it is different from an empty "" or a space ' '.**
 - NULL can be like "I didn't know there is a question".
 - Empty is consider a comment "I know the question and I have nothing to say".

Example 6.10 NULL Search Condition

- List details of all viewings on property PG4 where a comment has not been supplied.

SELECT clientNo, viewDate **FROM** dreamhome.Viewing
WHERE propertyNo = 'PG4'
AND comment **IS NULL**;

- Negated version (**IS NOT NULL**) can test for non-null values.

clientNo	propertyNo	viewDate	comment
CR56	PA14	2013-05-24	too small
CR56	PG36	2013-04-28	
CR56	PG4	2013-05-26	NULL
CR62	PA14	2013-04-14	no dining room
CR76	PG4	2013-04-20	too remote

clientNo	viewDate
CR56	26-May-04

NULL & Boolean in MySQL

- NULL is UNKNOWN, not FALSE, TRUE, or EMPTY
- Since MySQL doesn't have a boolean type, 1 is equivalent to TRUE, while 0 is equivalent to FALSE.

```
mysql> SELECT 0 IS NULL, 0 IS NOT NULL, '' IS NULL, '' IS NOT NULL;
```

0 IS NULL	0 IS NOT NULL	'' IS NULL	'' IS NOT NULL
0	1	0	1

```
mysql> select (NULL or 1), (NULL and 1), (NULL or 0), ( NULL and 0);
```

(NULL or 1)	(NULL and 1)	(NULL or 0)	(NULL and 0)
1	NULL	NULL	0

Class exercises – NULL and operations

- Use **dreamhome.Viewing**
 - List the clientno who didn't know there is a comment.
- Use **dreamhome.Staff1**
 - List the male staff names who have supervisors
- What are the following outputs?
NULL = NULL, NULL + 10
NULL AND 1, NULL AND 0
NULL OR 0, NULL OR 1

Example 6.11 Single Column Ordering

- List salaries for all staff, arranged in descending order of salary.

```
SELECT staffNo, fName,  
lName, salary  
FROM dreamhome.Staff  
ORDER BY salary DESC;
```

staffNo	fName	lName	salary
SL21	John	White	30000.00
SG5	Susan	Brand	24000.00
SG14	David	Ford	18000.00
SG37	Ann	Beech	12000.00
SA9	Mary	Howe	9000.00
SL41	Julie	Lee	9000.00

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003
SL21	John	White	Manager	M	1945-10-01	30000.00	B005
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005

Example 6.12 Multiple Column Ordering

- Produce abbreviated list of properties in order of property type.

SELECT propertyNo,
type, rooms, rent
FROM dreamhome.PropertyForRent
ORDER BY type;

propertyNo	type	rooms	rent
PL94	Flat	4	400
PG4	Flat	3	350
PG36	Flat	3	375
PG16	Flat	4	450
PA14	House	6	650
PG21	House	5	600

propertyNo	street	city	postcode	type	rooms	rent	ownerNo	staffNo	branchNo
PA14	16 Holh...	Aberdeen	AB7 5SU	House	6	650.00	CO46	SA9	B007
PG16	5 Novar ...	Glasgow	G12 9AX	Flat	4	450.00	CO93	SG14	B003
PG21	18 Dale ...	Glasgow	G12	House	5	600.00	CO87	SG37	B003
PG36	2 Manor...	Glasgow	G32 4QX	Flat	3	375.00	CO93	SG37	B003
PG4	6 Lawre...	Glasgow	G11 9QX	Flat	3	350.00	CO40	NULL	B003
PL94	6 Argyll ...	London	NW2	Flat	4	400.00	CO87	SL41	B005

Example 6.12 Multiple Column Ordering

- Four flats in this list - as no minor sort key specified, system arranges these rows in any order it chooses.
- To arrange in order of rent, specify minor order:

SELECT propertyNo, type,
rooms, rent
FROM dreamhome.PropertyForRent
ORDER BY type, rent **DESC**;

propertyNo	type	rooms	rent
PG16	Flat	4	450
PL94	Flat	4	400
PG36	Flat	3	375
PG4	Flat	3	350
PA14	House	6	650
PG21	House	5	600

propertyNo	street	city	postcode	type	rooms	rent	ownerNo	staffNo	branchNo
PA14	16 Holh...	Aberdeen	AB7 5SU	House	6	650.00	CO46	SA9	B007
PG16	5 Novar ...	Glasgow	G12 9AX	Flat	4	450.00	CO93	SG14	B003
PG21	18 Dale ...	Glasgow	G12	House	5	600.00	CO87	SG37	B003
PG36	2 Manor...	Glasgow	G32 4QX	Flat	3	375.00	CO93	SG37	B003
PG4	6 Lawre...	Glasgow	G11 9QX	Flat	3	350.00	CO40	NULL	B003
PL94	6 Argyll ...	London	NW2	Flat	4	400.00	CO87	SL41	B005

Class exercises – Sorting

- Use **dreamhome.Staff**
- List the Staff name, position, and salary.
- The output should be sorted:
 - first by position from a to z,
 - and then sorted by the salary from low to high.

Lab #1, question#5

- Follow the requirements to write the select statement, check the select statement returns the correct result.
- **create view vLab11_xxxx as select ..**
 - xxxx is your database login
- (1) switch to class database
 - use {class_database};**
- (2) The following example is for demo only, is not the answer
 - CREATE view vLab11_demo AS**
 - SELECT staffNo FROM dreamhome.Staff ;**
- (3) Verify the result:
 - select * from vLab11_demo;**
- (4) If the result is not correct, drop the view, then re-create view
 - (Q) Where to get the drop views permission? (A)Our class website.
 - syntax: drop view vLab11_demo;
- **Do NOT create the views until you learn how to create and drop views**
 - Please make sure the SELECT statement is working correctly before creating the view.

Homework & lab exercise #1

- Follow the requirements
- Write your SQL statements in your PC workspace first.
- Make sure your SQL statements product the correct results.
 - Manually verify your results

Homework & lab exercise #2

- **Important Note: Do not create views and tables until you learn how to get the permissions to create and drop them.**

- Login to class webpage
- Run option to get permission to select, create table/view/procedure on **class database**.
- Login to database server on Linux command line, and use the class database. For example:

```
mysql> Use CLASS_DATABASE_INSTANCE;
```

- You need to use the following statement: (xxxx is your email ID)

```
mysql> CREATE view vLab11_xxxx as SELECT .....
```

- For creating a table, you will need to use

```
mysql> CREATE table Test_xxxx
```

Home exercises

- SQL exercises questions 1 - 25
- Check tables
 - Staff, Branch, Customers
- Connect to the database
- Try the examples in textbook/slides
 - List all tables
 - SELECT all records and all columns
 - SELECT certain columns
 - SELECT specific records
 - Desc table
- SELECT with AS, ORDER BY, BETWEEN AND

Reading Assignment

- 6.2 SQL
- 6.3 DML – SELECT, DISTINCT, FROM, WHERE, LIKE, AS
- Exercises 6.1 – 6.10

History of SQL

- In 1974, D. Chamberlin (IBM San Jose Laboratory) defined language called 'Structured English Query Language' (SEQUEL).
- A revised version, SEQUEL/2, was defined in 1976 but name was subsequently changed to SQL for legal reasons.
- In late 70s, **ORACLE** appeared and was probably first commercial RDBMS based on SQL.
- In 1987, ANSI and ISO published an initial standard for SQL.
- In 1989, ISO published an addendum that defined an 'Integrity Enhancement Feature'.
- In 1992, first major revision to ISO standard occurred, referred to as SQL2 or SQL/92.
- In 1999, SQL:1999 was released with support for object-oriented data management.

Leetcode.com SQL exercise

- 181. Employees Earning More Than Their Managers
 - <https://leetcode.com/problems/employees-earning-more-than-their-managers/>
- Given the Employee table, write a SQL query that finds out employees who earn more than their managers. For the above table, Joe is the only employee who earns more than his manager.

Id	Name	Salary	ManagerId
1	Joe	70000	3
2	Henry	80000	4
3	Sam	60000	NULL
4	Max	90000	NULL

Employee
Joe

Output

Conclusion

- Exercise the SQL query on database sever
 - SQL: SELECT
- Test your database account
- Familiar with the commands on the Linux system and MySQL database.